1. What is an IP address?

□ Answer:

An IP address is a unique identifier assigned to a device in a network, used for communication between devices.

IPv4 is a 32-bit address used to identify devices on a network, represented in four decimal numbers separated by dots.

IPv6 is the **new version of IP address** created to solve IPv4 shortage.

IPv6 is a 128-bit address designed to provide a larger address space and improve network efficiency.

Feature	IPv4	IPv6
Size	32-bit	128-bit
Format	Decimal	Hexadecimal
Example	192.168.1.1	2001:db8::1
Address count	Limited	Very large
Speed	Slower	Faster

IPv4 Class IPv4 Start Address IPv4 End Address Usage A 0.0.0.0 127.255.255.255 Used fo Large Network B 128.0.0.0 191.255.255.255 Used fo Medium Size Network C 192.0.0.0 223.255.255.255 Used fo Local Ar Network D 224.0.0.0 239.255.255.255 Reserve for Multicas E 240.0.0.0 255.255.255.254 Study a R&D

2. What is DNS?

□ Answer:

DNS (Domain Name System) converts domain names into IP addresses so that the browser can connect to the correct server.

□ 3. How DNS works? (VERY IMPORTANT □)

□ Answer:

When a user enters a URL, the browser sends a request to the DNS server, which resolves the domain name into an IP address, and then the browser connects to that IP.

Answer:

HTTP is a protocol used for communication between client and server.

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5. Difference between HTTP and HTTPS?

Answer:

HTTP is not secure, while HTTPS is secure and uses encryption (SSL/TLS) to protect data.

7. What is TCP?

Answer:

TCP is a protocol that ensures reliable and ordered data transmission between devices.

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8. What is DHCP?

Answer:

DHCP automatically assigns IP addresses to devices in a network.

1. What is debugging?

Answer:

Debugging is the process of identifying and fixing errors in a system or application.

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2. Steps in debugging? (IMPORTANT)

Answer:

First understand the issue, reproduce it, check logs, identify the root cause, fix it, and test again.

1. What is a Port?

Answer:

A port is a communication endpoint used by applications to send and receive data. For example, HTTP uses port 80 and HTTPS uses port 443.

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2. What is a Firewall?

Answer:

A firewall is a security system that monitors and controls incoming and outgoing network traffic based on predefined rules.

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3. What is Ping?

Answer:

Ping is a command used to check connectivity between two devices and measure response time.

7. What is Subnet?

Answer:

A subnet is a smaller network created from a larger network to improve performance and security.

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8. What is MAC Address?

Answer:

A MAC address is a unique hardware identifier assigned to a network interface.

6. What is Gateway?

Answer:

A gateway is a device that connects different networks and allows communication between them.

Website loads but not fully

Answer:

I will check API responses, frontend errors, and network requests.

□ Slow API response

□ Answer:

I will check database queries, server load, and network latency.

□ Cannot access a website

□ Answer:

I will check DNS resolution, internet connection, and server availability.